



CLINOMA PLATFORM

PEDIATRICS

FLASHSPACE EDITION

Interactive High-Yield Visual Highlights & Explanations

EXCLUSIVE DIGITAL STUDY GUIDE

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TABLE OF CONTENTS

CHAPTER / CLINICAL SLIDE	PAGE
I. GROWTH AND DEVELOPMENT	P. 3
Growth charts	p. 4
Development milestones during 1st 4 years	p. 5
Key warning signs & delayed milestones	p. 6
II. NUTRITION	P. 7
Advantages of breast feeding & contraindication	p. 8
Nutritional disorders: Kwashiorkor & Marasmus	p. 9
Nutritional disorders: Rickets	p. 10
III. GIT	P. 11
Cow's milk allergy in Pediatrics	p. 12
GERD & Hypertrophic Pyloric Stenosis	p. 13
Vomiting in pediatrics	p. 14
Abdominal pain in pediatrics	p. 15
IV. GENETIC DISEASES	P. 16
Chromosomal disorders: Down Syndrome	p. 17
Chromosomal disorders: Turner syndrome	p. 18
Prenatal diagnosis (importance, indications & types)	p. 19

CHAPTER / CLINICAL SLIDE	PAGE
V. ENDOCRINOLOGY	P. 20
Short stature	p. 21
Hypothyroidism	p. 22
Diabetes Mellitus Type 1 & DKA	p. 23
Childhood obesity	p. 24
VI. HEMATOLOGY AND ONCOLOGY	P. 25
Iron deficiency anaemia	p. 26
Aplastic anaemia	p. 27
Chronic hemolytic anaemia	p. 28
RBC abnormalities: Spherocytosis	p. 29
RBC abnormalities: Thalassemias	p. 30
RBC abnormalities: G6PD deficiency	p. 31
Hemophilia	p. 32
ITP (Immune Thrombocytopenic Purpura)	p. 33
ALL & Prognostic factors	p. 34
Hodgkin lymphoma	p. 35

CHAPTER / CLINICAL SLIDE	PAGE
VII. CVS	P. 36
Acyanotic Heart: VSD (Part 1)	p. 37
Acyanotic Heart: VSD (Part 2)	p. 38
Acyanotic Heart: ASD	p. 39
Acyanotic Heart: PDA	p. 40
Cyanotic Heart: TGA	p. 41
Cyanotic Heart: Fallot / TOF	p. 42
Heart Failure	p. 43
VIII. CNS	P. 44
Cerebral Palsy	p. 45
Floppy infant	p. 46
Duchenne muscle dystrophy	p. 47
Infection	p. 48

Tip: Use the interactive landscape view on your tablet or laptop for optimal clinical visualization.

CHAPTER 01

I. Growth and development



GROWTH CHARTS • DEVELOPMENT MILESTONES DURING 1ST 4 YEARS • KEY WARNING SIGNS & DELAYED MILESTONES

[Image Missing: Growth charts.jpeg]

[Image Missing: Developmental milestones.jpeg]

[Image Missing: Key development warning signs & Delayed milestone causes.jpeg]

CHAPTER 02

II. Nutrition



ADVANTAGES OF BREAST FEEDING & CONTRAINDICATION • NUTRITIONAL DISORDERS: KWASHIORKOR & MARASMUS • NUTRITIONAL DISORDERS: RICKETS

[Image Missing: advantages of breastfeeding & contraindication.jpeg]

[Image Missing: PEM.jpeg]

[Image Missing: Rickets.jpeg]

CHAPTER 03

III. GIT



COW'S MILK ALLERGY IN PEDIATRICS • GERD & HYPERTROPHIC PYLORIC STENOSIS • VOMITING IN PEDIATRICS • ABDOMINAL PAIN IN PEDIATRICS

[Image Missing: Cow milk allergy.jpeg]

[Image Missing: GERD & HPS.jpeg]

[Image Missing: Vomiting.jpeg]

[Image Missing: Pediatrics abdominal pain.jpeg]

CHAPTER 04

IV. Genetic Diseases



CHROMOSOMAL DISORDERS: DOWN SYNDROME • CHROMOSOMAL DISORDERS: TURNER SYNDROME • PRENATAL DIAGNOSIS (IMPORTANCE, INDICATIONS & TYPES)

[Image Missing: Down syndrome.jpeg]

[Image Missing: Turner syndrome.jpeg]

[Image Missing: Prenatal diagnosis.jpeg]

CHAPTER 05

V. Endocrinology



SHORT STATURE • HYPOTHYROIDISM • DIABETES MELLITUS TYPE 1 & DKA • CHILDHOOD OBESITY

[Image Missing: Short stature.jpeg]

[Image Missing: Congenital hypothyroidism.jpeg]

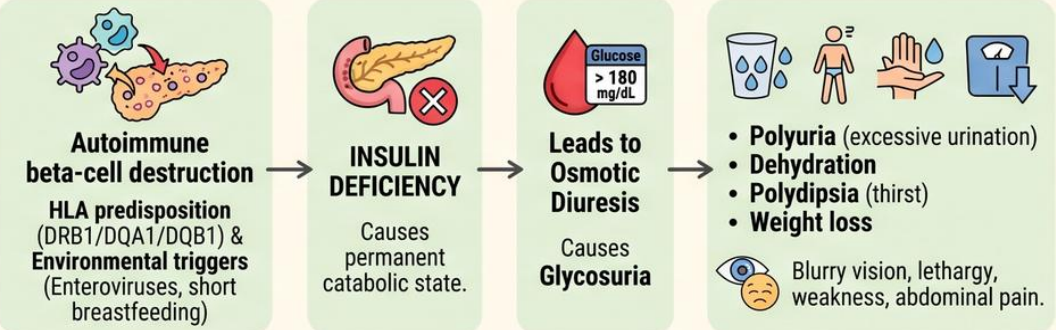
DIABETES MELLITUS (DM) & DIABETIC KETOACIDOSIS (DKA)

A Comprehensive Explanatory Overview

TYPE 1 vs TYPE 2 DM (Comparison)

TYPE 1	TYPE 2
 Sudden onset, children	 Gradual onset, adults
 Thin/normal body	 Often obese
 Ketoacidosis common	 Ketoacidosis rare
 Autoantibodies present (ICA, GAD-65, IAA, IA-2A)	 Autoantibodies absent
 Low/absent endogenous insulin	 Normal/increased insulin
 50% twin concordance	 90% twin concordance Genetic link stronger in T2

TYPE 1 DM PATHOGENESIS & PRESENTATION (Core Mechanism)



DIABETIC KETOACIDOSIS / DKA (Medical Emergency)



LIFE-THREATENING MEDICAL EMERGENCY with 10% MORTALITY
Precipitated by: Infection (UTI), Negligence/Omission, Trauma, Vomiting.

DIAGNOSIS & INSULIN THERAPY (Management & Criteria)

DIAGNOSIS CRITERIA



- HbA1C > 6.5%
- FPG > 126 mg/dL
- Random glucose > 200 mg/dL (with symptoms)

INSULIN THERAPY MANAGEMENT



Lines of Treatment: Diet, Exercise, Glucose Monitoring & Insulin



BOLUS (Meal):
Rapid/Short-acting (Lispro, Aspart) – 15 mins;
(Regular) – 30–60 mins before meals.



BASAL (Background):
Intermediate (NPH, Lente) – 10–16 hours;
Long-acting (Glargine, Detemir) – 20–24 hours.



Pre-mixed NOT recommended in pediatrics.

IMMUNE MARKERS & C-PEPTIDE



Autoantibodies:
GAD-65, ICA, IAA, IA-2A.

C-peptide < 0.2 nmol/l
confirms Type 1

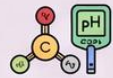
DIAGNOSTIC TRIAD



HYPERGLYCEMIA
(> 300 mg%)



KETONEMIA/KETONURIA



ACIDOSIS
(pH < 7.35, HCO₃ < 16 mEq/L)

CLINICAL PRESENTATION & SEVERITY



Signs: Kussmaul respiration (deep, rapid), Acetone breath, Abdominal pain, Dehydration (dry tongue, sunken eyes), Coma.

Mild

Mod

Severe

SEVERE: Venous pH < 7.15; HCO₃ < 10 mEq/L; depressed respirations/coma.

MANAGEMENT PROTOCOL



- Immediate Hospitalization
- IV Fluids Correction
- Electrolyte Correction (esp. Potassium)
- Acid-base Balance
- Low-dose Continuous Insulin
- Treat Precipitating Causes.

DEFINITION & PERCENTILES

CHILDHOOD OBESITY

Obesity is a major pediatric health issue linked to childhood complications and lifelong risks.



Obesity/increased adiposity is defined using **BODY MASS INDEX (BMI)**, an excellent proxy for body fat.



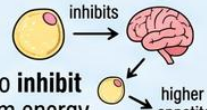
$$\text{BMI} = \frac{\text{weight in kg}}{\text{height in meters}^2}$$

Other measures: waist circumference (WC), hip circumference (HC), waist-to-hip ratio (WHR).

Pediatric Obesity Definition (Over 2 years, BMI charts)



Leptin Hormone Role: **LEPTIN**, released from fat cells, signals hypothalamus to **inhibit hunger** and regulate long-term energy. **Leptin levels connect directly to body fat amount.** Weight loss reduces leptin, potentially triggering appetite increases, making weight loss difficult.



COMPLICATIONS & DIAGNOSIS

Physical Complications



Type 2 Diabetes: Increased by obesity and sedentary lifestyle.



Cardiovascular Risk: High Cholesterol, High Blood Pressure, Metabolic Syndrome, dyslipidemia



Joint Pain: Hips, knees, and back pain/injuries due to extra weight



Breathing Problems: Asthma, Obstructive Sleep Apnea



NAFLD: Non-alcoholic Fatty Liver Disease, liver damage from fatty deposits

Social/Emotional Complications

Bullying, loss of self-esteem, depression, anxiety

Diagnosis & Evaluation



Clinical Assessment: Family history (nutritional/activity habits), pediatric exam for cause and comorbidities.



Laboratory Investigations: Transaminases (AST, ALT), basal glycaemia, lipid profile. **HOMA Index** (fasting glucose x insulin / 22.5) indicates **Insulin Resistance (IR)**, even with normal fasting glycaemia. Baseline screening for GH, IGF-I, IGFBP-3, iPTH, thyroxine, TSH, electrolytes (for hypercortisolism/pseudohypoparathyroidism).



OGTT: Oral-Glucose-Tolerance Test if family history of T2D or IR signs like Acanthosis nigricans/PCOS.



Imaging:

- Bone Age Radiograph** (Left hand/wrist): Shows **accelerated bone age** vs chronological age
- Abdominal Ultrasound:** If non-alcoholic steatohepatitis or PCOS is suspected

ETIOLOGY & RISK FACTORS

Complex Interaction: Obesity is a complex disease with multiple interacting factors increasing risk.

Environmental & Diet:

- Lifestyle:** Too little activity, too many calories.
- Diet:** High-calorie **fast foods**, baked goods, snacks, candy, **sugary drinks** (fruit juices, sports drinks) cause weight gain.



Lack of Exercise: Sedentary activities (TV, video games) burn fewer calories.



Family Factors: Family environment with available high-calorie foods and lack of encouraged physical activity.



Psychological: Stress (personal, parental, family) can lead to overeating for coping, emotions, or boredom.



Socioeconomic: Limited resources lead to convenience foods (frozen meals) and lack of safe exercise places.



Hormonal Factors: Underlying endocrinopathies like **Cushing syndrome** and **hypothyroidism**.



Genetic Factors: Syndromes such as **Prader-Willi Syndrome**.



Certain Medications: Prescription drugs like **prednisone**, **lithium**, **propranolol**.

DIFFERENTIAL DIAGNOSIS, PREVENTION & TREATMENT

Differential Diagnosis (High-Yield Contrast)



Nutritional Obesity: Consistent/accelerated growth, early puberty, advanced **bone age** (> 2 SD).



Endocrine Obesity: Decreased/decelerated linear growth. Test for thyroid hormones.



Genetic Syndrome Obesity: Severe obesity < 5 years. Developmental delay, short stature, dysmorphic facies, hyperphagia.



Screening: **DNA methylation**, exome sequencing, karyotyping for syndromes (Prader-Willi, etc.)

Prevention

Healthy family lifestyle (eating/activity), low-fat healthy snacks, expose to new foods, **non-food rewards**, adequate sleep



Treatment Lines

1. **Diet Control:** Appropriate calories (fruits, vegetables, whole grains, lean meat, fish, poultry). Avoid bad habits (eating in bedroom).



2. **Increasing Physical Activity:** Walking to school, leisure family activities, organized sports.



3. **Pharmacotherapy:** No effective reversal drug. FDA-approved: **ORLISTAT** for children < 16 years, decreases fat absorption, modest weight loss.



4. **Bariatric Surgery:** STRICT CRITERIA:

Complete/near-complete **skeletal maturity**, **BMI ≥ 40**, medical complication present, **FAILED ≥ 6 months** organized weight management



CHAPTER 06

VI. Hematology and oncology

IRON DEFICIENCY ANAEMIA • APLASTIC ANAEMIA • CHRONIC HEMOLYTIC ANAEMIA • RBC ABNORMALITIES: SPHEROCYTOSIS • RBC ABNORMALITIES: THALASSEMIAS • RBC ABNORMALITIES: G6PD DEFICIENCY • HEMOPHILIA • ITP (IMMUNE THROMBOCYTOPENIC PURPURA) • ALL & PROGNOSTIC FACTORS • HODGKIN LYMPHOMA

[Image Missing: Iron defeciency anemia.jpeg]

[Image Missing: Aplastic anemia.jpeg]

CHRONIC HEMOLYTIC ANEMIA & HEREDITARY SPHEROCYTOSIS: CLINICAL & PATHOPHYSIOLOGICAL OVERVIEW

1. The Classical Clinical Triad & Acute Crises

The Hemolytic Triad: Clinically Characterized

ANEMIA:



Presenting as progressive **skin & mucosal pallor**, **fatigue**, and **exercise intolerance**.

JAUNDICE:



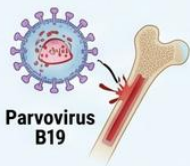
Manifesting as **scleral icterus**, accompanied by **dark-colored urine (bilirubinuria)**.

HEPATOSPLENOMEGALY (HSM):



Splenic enlargement driven by **chronic overwork** **clearing defective cells**, paired with variable liver enlargement.

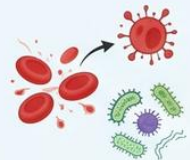
Life-Threatening Acute Crises: (Dangerous!)



1. APLASTIC CRISIS:

MOST DANGEROUS!

Almost universally triggered by **Parvovirus B19**. Completely **halts marrow erythropoiesis**. Sudden drop in hemoglobin & **profound reticulocytopenia (<1%)**.



2. HEMOLYTIC CRISIS:

Sudden, temporary **acceleration** of RBC destruction, precipitated by **acute bacterial or viral infections**.

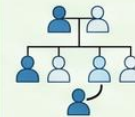


3. MEGALOBLASTIC CRISIS:

Due to **rapid, severe depletion of nutritional folate stores** driven by chronic, highly **accelerated bone marrow erythropoiesis**.

2. Hereditary Spherocytosis (HS): Pathophysiology

Definition & Genetics: Common Inherited Defect



MOST COMMON INHERITED RBC MEMBRANE DEFECT:

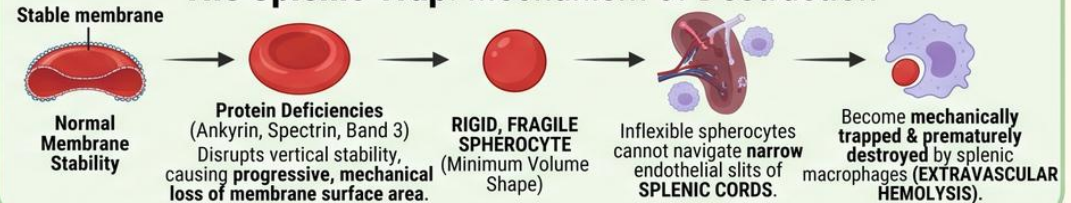
Primarily transmitted as an **AUTOSOMAL DOMINANT TRAIT**.



MOLECULAR TRAIT:

Caused by **genetic mutations** affecting vital structural **membrane proteins**: most frequently **ANKYRIN, SPECTRIN, or BAND 3**.

The Splenic Trap: Mechanism of Destruction



3. Clinical Presentation & Laboratory Diagnostics

Clinical Manifestations: Key Symptoms



NEONATAL JAUNDICE

Prominent, within the **first few days of life**.



MILD-TO-MODERATE CHRONIC ANEMIA

Ongoing condition.



PROGRESSIVE SPLENOMEGALY

Spleen enlargement over time.



HIGH RISK OF PIGMENT GALLSTONES

(Calcium bilirubinate cholelithiasis) in **early childhood** due to chronic hyperbilirubinemia.

Laboratory Diagnostics: Key Findings



COMPLETE BLOOD COUNT (CBC)

Normocytic/mild microcytic anemia. **Elevated MCHC (>36 g/dL)** (Key Signature: cellular dehydration). **Reticulocytosis (>2%)** (Bone marrow compensation).



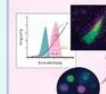
OSMOTIC FRAGILITY TEST

Demonstrates **significantly increased RBC lysis** in hypotonic saline due to **reduced surface-to-volume ratio**.



PERIPHERAL BLOOD SMEAR (PBS)

Directly confirms numerous **SPHEROCYTES**: small, dark, hyperchromic round RBCs **lacking central pallor**.



EMA BINDING TEST

(Flow Cytometry) **MODERN GOLD STANDARD**. Quantifies marked **reduction in fluorescent intensity bound to membrane proteins**. High sensitivity & specificity.

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[Image Missing: Thalassemia.jpeg]

[Image Missing: GP6D.jpeg]

[Image Missing: Hemophilia.jpeg]

[Image Missing: Immune thrombocytopenia (ITP).jpeg]

[Image Missing: Acute lymphoplastic leukemia.jpeg]

[Image Missing: HODGKIN lymphoma.jpeg]

CHAPTER 07

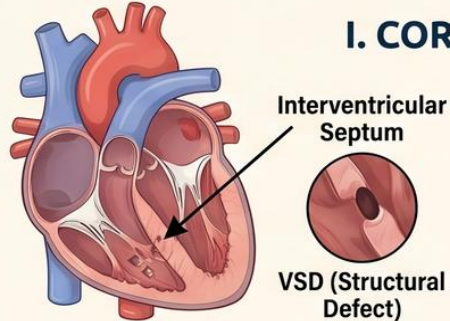
VII. CVS



ACYANOTIC HEART: VSD (PART 1) • ACYANOTIC HEART: VSD (PART 2) • ACYANOTIC HEART: ASD • ACYANOTIC HEART: PDA • CYANOTIC HEART: TGA • CYANOTIC HEART: FALLOT / TOF
• HEART FAILURE

VENTRICULAR SEPTAL DEFECT (VSD):

PART 1 - Anatomy & Hemodynamics



I. CORE DEFINITION & PREVALENCE

Keypoint 1:
Concept: A **structural defect** in the **interventricular septum**.

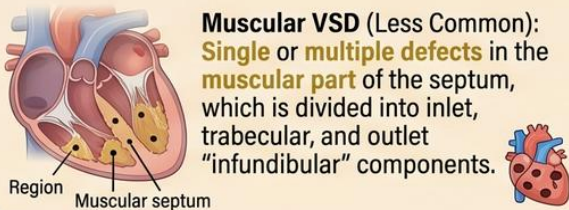
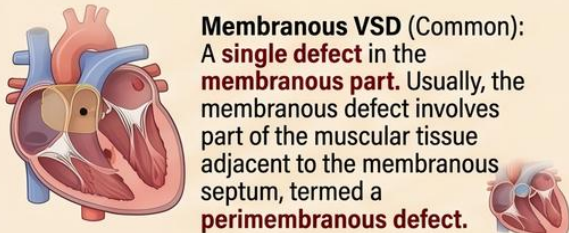


Keypoint 2:
Prevalence: VSD is the **most common form** of congenital heart disease (CHD) and accounts for about **30% of all such defects**.

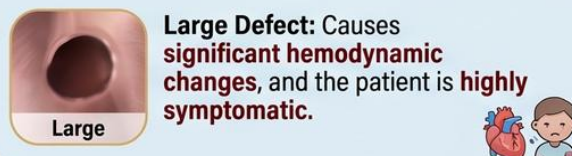
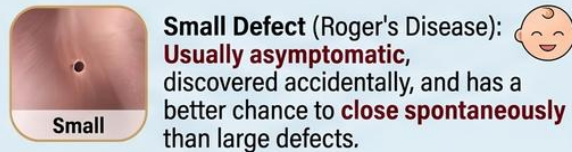


II. ANATOMICAL AND SIZE CLASSIFICATION

A. According to the Site of the Defect

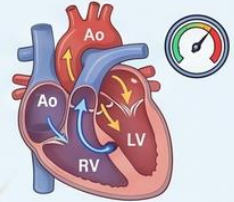


B. According to the Size of the Defect

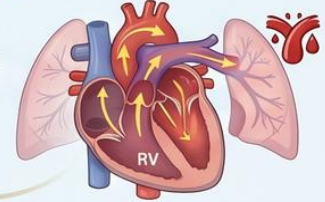


III. COMPLEX HEMODYNAMICS

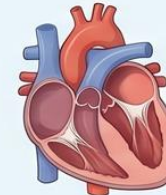
Keypoint 1:
Shunt Mechanism: Blood from the **left ventricle (LV)** is **divided into 2 portions**: one portion passes to the aorta and the other passes **through the defect to the right ventricle (RV)**. The amount of shunted blood depends on the **size of the VSD** and the level of **pulmonary vascular resistance**.



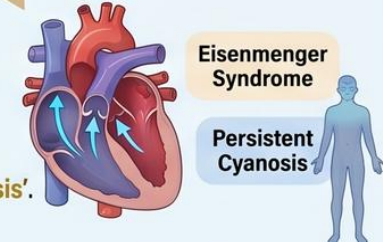
Keypoint 2:
Pulmonary Plethora: **All this blood** (RV output & part from LV output that came through VSD) passes through the **pulmonary artery**, leading to **pulmonary plethora**.



Keypoint 3:
Chamber Dilatation: This blood passes to the **left atrium (LA)**, leading to **left atrium dilatation** and subsequent **left ventricular hypertrophy (LVH)** or **left ventricular dilatation**.



Eisenmenger Syndrome: Persistence of pulmonary plethora causes **pulmonary hypertension**, leading to symptoms of **low cardiac output (COP)**. Pulmonary hypertension **increases right ventricular pressure**, causing Eisenmenger Syndrome and resulting in '**persistent cyanosis**'.



VENTRICULAR SEPTAL DEFECT (VSD): CLINICAL PICTURE & MANAGEMENT (PART 2)

IV. CLINICAL MANIFESTATIONS

A. Symptoms



Small VSD: Asymptomatic, normal growth & development.



Moderate to Large VSD: Delayed growth, tachypnea & excessive sweating (feeding), repeated pulmonary infections, CHF symptoms (infancy). Exertional dyspnea (older children).



Eisenmenger's Phase: Cyanosis, clubbing, decreased activity (with pulmonary vascular obstructive disease).

B. Signs



Inspection & Palpation: Precordial bulge, hyperkinetic apex, systolic thrill (lower left sternal border) with large VSD.



Auscultation (Heart Sounds): P2 intensity normal (small shunt) / increased (large shunt). S2 loud & single (pulmonary hypertension).



Auscultation (Murmurs): Harsh pansystolic murmur (lower left sternal border, propagated). Sometimes early systolic. **Infundibular VSD:** soft early diastolic AR murmur (aortic cusp herniation).

V. INVESTIGATIONS



Electrocardiography (ECG): Normal (small). LVH & LAH (moderate). BVH +/- LAH (large). RVH only (pulmonary vascular obstructive disease).



Radiography (CXR): Cardiomegaly (LA, LV, sometimes RV). Increased pulmonary vascular markings. Relates to shunt magnitude.

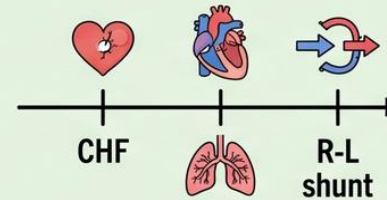


Echocardiography: Identifies defect (number, size, location), estimates PA pressure, detects chamber enlargement, associated defects, shunt magnitude.

VI. NATURAL HISTORY



Spontaneous Closure: More frequent with small defects & first 6 months. ~60% muscular (by 8 yrs), ~35% perimembranous (by 5 yrs). Inlet/outlet do not close.



Complications Timeline: CHF (large VSDs, 6-8 weeks). Pulmonary vascular obstructive disease (large VSDs, as early as 6-12 months, R-L shunt usually teenage years).

VII. MANAGEMENT



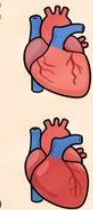
Medical Management: Diuretics & ACE inhibitors (CHF, 2-4 months for growth). High-calorie feedings (NG tube/oral). No exercise restriction (no hypertension).



Definitive Closure Indications & Timing: Failure to thrive. PA pressure >50% systemic (by 1 year). Dilated left chambers, intractable CHF, subaortic VSD with AR.



Procedures: Nonsurgical device closure (selected muscular). PA banding (palliative, rarely used). Surgical closure (patch closure under bypass).



ATRIAL SEPTAL DEFECT (ASD)

I. DEFINITION & PREVALENCE

Concept: Structural defect in the inter-atrial septum.

Prevalence

Isolated ostium secundum defect occurs in **5% to 10%** of all congenital heart defects (CHDs).

Gender Ratio: Significantly **more common** in **females** than in males, with a male-to-female ratio of 1:2.

II. ANATOMICAL TYPES

Ostium Secundum Defect

Commonest presentation, located in the **center** of the septum.

Ostium Primum Defect

Located in the **lower portion** of the septum.

Sinus Venosus Defect

Located high up near **superior vena cava entry**, or lower down near **inferior vena cava**.

Coronary Sinus ASD

Located within the **roof of the coronary sinus**.

III. HEMODYNAMICS

Shunt Mechanism

Left atrial pressure is **physiologically higher** than right atrial pressure. Therefore, blood is **continuously shunted** from the left atrium (LA) to the right atrium (RA).

Volume Overload

The RA receives extra shunted volume and normal venous return, leading directly to **right atrial dilatation and enlargement**.

Pulmonary Consequences

Large volume propelled into **right ventricle (RV)**, causing marked **RV volume overload and dilatation**. Blood exits to **pulmonary artery**, producing **pulmonary artery enlargement**, lung plethora, and repeated chest infections.

IV. CLINICAL MANIFESTATIONS

A. Symptoms

Small ASD

Infants **well-developed** and remain **completely asymptomatic**.

Large ASD

Infants may present with **poor weight gain** or show **progressive symptoms and signs of CHF**.

B. Physical Signs

Inspection & Palpation

Large left-to-right shunt: **visible precordial bulge**, **prominent RV cardiac impulse**, **palpable PA pulsations**.

Heart Sounds

The second heart sound (S_2) is **widely split and fixed**, which stands as the **pathognomonic and characteristic finding** of ASD.

Murmurs

Soft (grade 2 to 3/6) **systolic ejection murmur** best audible over pulmonary area (functional murmur due to relative pulmonary stenosis). With large shunt: **mid-diastolic rumble** at lower left sternal border (relative tricuspid stenosis).

Crucial Note

There is absolutely **NO MURMUR** produced by the actual passage of blood through the ASD itself.

V. INVESTIGATIONS

Electrocardiography (ECG)

Typical findings show **right axis deviation (RAD)** accompanied by **mild RVH** or a **right bundle branch block (RBBB)** pattern in lead V_1 .

Chest Radiography (CXR)

Reveals **distinct cardiomegaly** with clear **enlargement of RA and RV**. **Prominent pulmonary artery (PA) segment** and **increased pulmonary vascular markings** are visible with significant shunt.

V. INVESTIGATIONS

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Typical findings show **right axis deviation (RAD)** accompanied by **mild RVH** or a **right bundle branch block (RBBB)** pattern in lead V_1 .

Chest Radiography (CXR)

Reveals **distinct cardiomegaly** with clear **enlargement of RA and RV**. **Prominent pulmonary artery (PA) segment** and **increased pulmonary vascular markings** are visible with significant shunt.

Echocardiography

Definitive diagnostic modality; identifies exact **anatomical type and size**, estimates **PA pressure**, detects **chamber enlargement**, rules out associated defects, and estimates **shunt magnitude (Q_p/Q_s)**.

VI. MANAGEMENT

Exercise Guidelines

Absolutely **no exercise restriction** is necessary.

Medical Management

Recommended in infants with **CHF** due to high success rate and possibility of **spontaneous defect closure**.

Definitive Closure Timing

Considering spontaneous closure, ASDs are preferably closed when children reach **2-4 years**.

Early Closure Indications

Closure indicated **before 2 years** if:

- Significant left-to-right shunt with **clear evidence of RV volume overload** (dilated right chambers or $Q_p/Q_s \geq 1.5:1$);
- Failure to thrive (FTT)**;
- Congestive heart failure** completely unresponsive to medical management.

Nonsurgical Device Closure

Transcatheter closure using catheter-delivered device is the **absolute standard preferred method**, applicable only for **secundum ASD**.

Surgical Closure

Strictly reserved for patients where device closure is **not anatomically appropriate**. Performed via **simple suture or patch closure** under cardboard bypass.

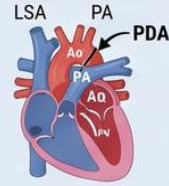
PATENT DUCTUS ARTERIOSUS (PDA)

EXPLANATORY GUIDE

I. DEFINITION & PREVALENCE

Concept

It is the **persistence** of the normal fetal communication between the **main pulmonary artery** and the **aortic arch**, located specifically **after the origin of the left subclavian artery**.



Prevalence

Occurs in **5% to 10%** of all CHDs when excluding premature infants.



Gender Ratio

It is significantly more common in females than in males, carrying a **male-to-female ratio of 1:3**. It is a highly common and major clinical problem in **premature infants**.



II. HEMODYNAMICS

Shunt Mechanism

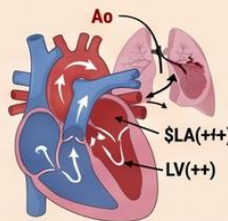
Pressure in the systemic aorta remains **higher** than the pressure in the pulmonary artery during both systole and diastole.



Chamber Overload

Consequently, blood is continuously shunted from the aorta into the pulmonary artery, which simultaneously receives normal venous output from the RV.

This leads directly to **increased pulmonary blood flow**, lung plethora, **left atrial dilatation (LA++)**, and **left ventricular dilatation/hypertrophy (LV++)**, creating a physiological circuit very similar to VSD.



III. CLINICAL MANIFESTATIONS

A. Symptoms



Small PDA

The patient is completely **asymptomatic**.



Large PDA

Frequently causes **poor weight gain**, decreased exercise tolerance, **repeated lower respiratory tract infections**, and **congestive heart failure (CHF)** during. Older children may present with **exertional dyspnea**.



B. Physical Signs

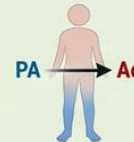
General Examination

Infants with secondary CHF show prominent **tachycardia** and **tachypnea**. Large PDAs present with **bounding peripheral pulses** and a classic **water hammer pulse**, caused directly by a wide pulse pressure with elevated systolic pressure and lower diastolic pressure.



Differential Cyanosis Risk

If pulmonary vascular obstructive disease develops, it induces a **right-to-left ductal shunt**, resulting in **cyanosis strictly in the lower half of the body** (differential cyanosis).



Inspection & Palpation

Large shunts present with a **hyperactive precordium**. A **palpable systolic thrill** may be present at the **upper left sternal border** in the left infraclavicular area.



Auscultation (Heart Sounds)

P_2 is usually normal but its intensity becomes **markedly accentuated** if pulmonary hypertension is present.



Auscultation (Murmur)

Characterized by a pathognomonic grade 1 to 4 of 6 **continuous machinery murmur**, best audible at the **left infraclavicular area** and propagated to the pulmonary area and the back.



IV. INVESTIGATIONS

Electrocardiography (ECG)

Findings are highly similar to VSD. A normal ECG or **isolated LVH** is seen with a small-to-moderate PDA, whereas **biventricular hypertrophy (BVH)** is characteristically seen with large PDAs.



Chest Radiography (CXR)

Radiographs are similar to VSD. They may be completely normal with a small-shunt PDA. Moderate-to-large shunts reveal varying degrees of **cardiomegaly** with specific enlargement of the **LA, LV, and ascending aorta**, accompanied by **increased pulmonary vascular markings**. If pulmonary vascular obstructive disease develops, heart size normalizes but displays a marked prominence of the **PA segment and hilar vessels**.



Echocardiography

Assesses the **exact size, site, and shape** of the PDA, and documents the **direction of flow** through the shunt. The measured dimensions of the LA and LV provide an indirect, objective assessment of the **magnitude of the left-to-right ductal shunt**.



V. NATURAL HISTORY & MANAGEMENT

B. Management Protocol

Medical Prophylaxis (Premature Only)

Pharmacological closure using **indomethacin, ibuprofen, or paracetamol** is highly effective and widely used to close the PDA **strictly in premature infants**.

Exercise Rules

No exercise restriction is needed in the absence of pulmonary hypertension.

Definitive Closure Indication

Establishing a **diagnosis of PDA** in a term infant is a **direct indication for its closure**, except in a completely silent PDA with no hemodynamic consequences.

Nonsurgical Intervention

Nonsurgical device occlusion of the PDA via **cardiac catheterization** has become the **standard of care**.

Surgical Intervention

Surgical closure (ligation or patch) is reserved **exclusively** for patients in whom device closure is not applicable.



COMPLETE TRANSPOSITION OF THE GREAT ARTERIES (TGA)

I. PREVALENCE & CORE DEFINITION

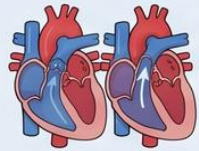


PREVALENCE
Most common **cyanotic** disorder in neonates

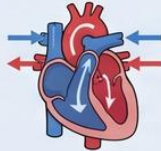
GENDER DISPARITY
More frequent in **males**
MALE-TO-FEMALE RATIO: 3:1



ANATOMICAL REVERSAL
Reversal of normal anteroposterior relation



NORMAL
Aorta **POSTERIOR & RIGHTWARD**
→ **ANTERIOR & LEFTWARD**



TGA
Aorta **ANTERIOR & RIGHTWARD**
→ **POSTERIOR & LEFTWARD**

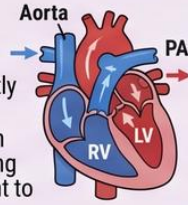
II. PATHOPHYSIOLOGY & PARALLEL CIRCULATIONS

INDEPENDENT CIRCUITS
Two independent circulations in **PARALLEL**, not series



ABNORMAL ORIGINS

- **Aorta** arises from **RIGHT VENTRICLE (RV)**, carrying **desaturated (blue)** blood directly to the body
- **Pulmonary Artery** arises from **LEFT VENTRICLE (LV)**, carrying **oxygenated (red)** blood straight to the lungs



SURVIVAL OBLIGATION
INCOMPATIBLE WITH LIFE without mixing

MIXING SITES
Essential for survival via **PFO, ASD, VSD, or PDA**



ANATOMICAL SUBTYPES
60% Intact Ventricular Septum (Atrial shunt only)



ANATOMICAL SUBTYPES

- **60% Intact Ventricular Septum** (Atrial shunt only)
- **40% VSD Present**

III. CLINICAL MANIFESTATIONS

A. HISTORY & SYMPTOMS



CYANOSIS TIMELINE
Intense **cyanosis since birth**



NEONATAL DISTRESS
Rapid CHF symptoms, progressive dyspnea, severe feeding difficulties



SEPTAL INFLUENCE
Intact Septum = Symptoms within **2 DAYS**, more intense cyanosis than VSD cases



B. PHYSICAL EXAMINATION SIGNS



CYANOSIS QUALITY
NOT RELIEVED by 100% O₂ therapy



AUSCULTATION (MURMURS)

- **Intact Septum:** No murmur or soft, non-specific
- **Associated VSD:** Classic **HARSH MURMUR**
- **Associated PS:** **DISTINCTIVE EJECTION SYSTOLIC MURMUR**



SYSTEMIC CONGESTION
Noticeable **HEPATOMEGALY**, worsening dyspnea (if CHF)

IV. DIAGNOSTIC INVESTIGATIONS



ELECTROCARDIOGRAM (ECG):

- Shows **RIGHT-AXIS DEVIATION (RAD)** and
- **RIGHT VENTRICULAR HYPERTROPHY (RVH)** due to systemic RV pressure

CHEST X-RAY (CXR):

- Generalized **CARDIOMEGALY**
- Increased **PULMONARY VASCULAR MARKINGS**
- Classic **EGG-SHAPED** ("EGG ON A STRING")



ECHOCARDIOGRAM
DEFINITIVE, PRIMARY DIAGNOSTIC MODALITY

V. MANAGEMENT PROTOCOL

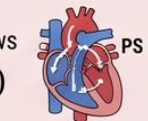
A. PALLIATIVE EMERGENCY PROCEDURES

- Improve mixing to prevent death from severe hypoxia & metabolic acidosis
- **INTRAVENOUS PROSTAGLANDIN (\$PGE_1)**:
- **IMMEDIATE** administration to keep **DUCTUS ARTERIOSUS PATENT**
- **RASHKIND BALLOON ATRIAL SEPTOSTOMY**
- Creates or enlarges **ATRIAL SEPTAL DEFECT** for mixing



B. DEFINITIVE CORRECTIVE OPERATIONS

- **ARTERIAL SWITCH OPERATION (ASO):**
- **PROCEDURE OF CHOICE**
Surgically switches Aorta and Pulmonary Artery to correct ventricles. Must occur **EARLY (3-8 WEEKS)**
- **ATRIAL (VENOUS) SWITCH**
Historical (Mustard/Senning) redirects venous inflows
- **VENTRICULAR SWITCH (RASTELLI OPERATION)**
For **COMPLEX PATIENTS (VSD + PS)**



TETRALOGY OF FALLOT (TOF) & HYPERCYANOTIC SPELLS (SUMMARY)

I. TOF CORE PATHOPHYSIOLOGY

Definition: Most common cyanotic disorder in children.

The 4 Classic Components:

1. Ventricular Septal Defect (VSD).
2. Overriding Aorta.
3. Pulmonary Stenosis (mainly infundibular).
4. Right Ventricular Hypertrophy (RVH).



Shunt Dynamics:

Right-to-left shunt via VSD causes systemic cyanosis. It varies dynamically with **exercise, emotion, or infundibular muscle spasms.**

II. CLINICAL MANIFESTATIONS

Cyanosis: Presents early, worsening after PDA closure, and **fails to respond to oxygen therapy.**



The Squatting Mechanism:

Older children squat to **increase systemic vascular resistance**, which reduces the right-to-left shunt and improves pulmonary blood flow.



Heart Failure Absence:

Congestive heart failure typically does not occur in TOF.



Auscultation Murmur:

A **systolic ejection murmur** at the left sternal border caused strictly by the **pulmonary stenosis**, not the VSD. The murmur **softens or disappears during a cyanotic spell.**



III. HYPERCYANOTIC (TET) SPELLS

Clinical Picture: Sudden attacks of deepening cyanosis, **tachypnea, irritability, and crying.** Severe spells can lead to **unconsciousness, convulsions, or death.**



Precipitating Factors:

Crying, defecation, fever, feeding, or awakening from naps. Iron deficiency can also trigger it.



Auscultation Murmur:

A **systolic ejection murmur** at the left sternal border caused strictly by the **pulmonary stenosis**, not the VSD. The murmur **softens or disappears** during a cyanotic spell.



IV. DIAGNOSTIC INVESTIGATIONS

ECG:

Shows **right-axis deviation (RAD)** and **right ventricular hypertrophy (RVH).**



Chest X-ray (CXR):

Characteristically reveals a normal heart size with a **boot-shaped contour ("coeur en sabot")** and decreased pulmonary vascular markings.



Echocardiography:

Stands as the **definitive, primary diagnostic modality.**



V. EMERGENCY MANAGEMENT PROTOCOL (TET SPELLS)

Execute these clinical steps immediately and strictly in order:



1. **Knee-Chest Position.**



2. **Calm the Child** (avoid iatrogenic agitation).



3. **Supplemental Oxygen** (via least aggravating method).



4. **Morphine Sulfate** (subcutaneous or slow IV).



5. **Sodium Bicarbonate IV** (to treat acidosis).



6. **Warmed Fluid Bolus** (\$10 ml/kg of 0.9% NaCl IV).



7. **Beta-Blockers** (e.g., Esmolol to reduce infundibular spasm).



8. **Ketamine IV or Phenylephrine IV** (to increase).



9. **Intubation / General Anesthesia.**



10. **Urgent Surgical Shunt.**



CRITICAL ABSOLUTE CONTRAINDICATIONS:

Give **NO INOTROPES** and **NO DIURETICS** during a spell.

VI. SURGICAL TREATMENT

Palliative:
Modified Blalock-Taussig shunt.



Definitive:
Complete surgical repair, preferred at 6 to 9 months of age.



PEDIATRIC HEART FAILURE (HF)



Definition: A clinical state where the heart cannot deliver an adequate cardiac output to meet the body's metabolic needs.

I. PATHOGENESIS & ETIOLOGY



Core Mechanisms & Causes



Increased Afterload (Pressure Overload): Seen in Aortic Stenosis (AS), Coarctation of the Aorta (CoA), systemic hypertension.



Increased Preload (Volume Overload): Leads to High-Output Failure. Seen in Left-to-Right shunts (VSD, PDA, ASD), valve regurgitation, severe anemia.



Decreased Myocardial Contractility: Seen in Cardiomyopathy, Myocarditis.



Tachyarrhythmias: Shorten diastolic filling time.



Precipitating Factors: Infections (endocarditis, pneumonia), physical/emotional stress, arrhythmias, anemia, iatrogenic factors (excessive IV fluids, beta-blockers in acute stages, corticosteroids causing fluid retention).

II. CLINICAL PICTURE



A. In Infants (The Diagnostic Triad)

The Essential Triad

Tachypnea Tachycardia Enlarged tender liver



Other Manifestations: Poor weight gain, restlessness, irritability, weak cry, feeding difficulties.

B. In Older Children

Left-Sided Heart Failure (Pulmonary Congestion & Low COP):



Low Cardiac Output (COP): Headache, syncope, pale cold skin, peripheral cyanosis, easy fatigability.



Pulmonary Congestion: Dyspnea, cough, hemoptysis, recurrent chest infections, bilateral basal crepitations, pleural effusion.

Cardiovascular Signs: Weak pulse, tachycardia, pulsus alternans, low systolic BP, apex beat shifted out & down (LV enlargement).



Right-Sided Heart Failure (Systemic Congestion):



Systemic Congestion: Congested pulsating neck veins, epigastric/right hypochondrium pain, enlarged tender liver, anorexia, nausea, vomiting, ascites. Lower limb edema is a late sign.



Cardiovascular Signs: Weak & rapid pulse, low systolic BP, apex shifted outwards (RV enlargement).

III. FUNCTIONAL CLASSIFICATION



Ross Classification (For Infants & Children):

Class I No limitations or symptoms.



Class II Mild tachypnea/diaphoresis with feedings; dyspnea on exertion in older children; no growth failure.



Class III Marked tachypnea/diaphoresis with feedings or exertion; prolonged feeding times with growth failure.



Class IV Symptomatic at rest with tachypnea, retractions, grunting, or diaphoresis.



IV. INVESTIGATIONS



Chest X-ray (CXR):

Cardiomegaly, signs of pulmonary edema (fluffy markings), increased pulmonary vascularity with Left-to-Right shunts.



ECG: Chamber dilatation, flat/inverted ST segment/T waves, causative arrhythmias.



Echocardiography:

Primary tool used to definitively assess ventricular function and identify structural causes.

V. GENERAL & NUTRITIONAL MANAGEMENT



Positioning & Oxygen:

Keep patient in a semi-upright position, provide humidified oxygen (40%–50%).

Infant Hypercaloric Rules:

Infants need significantly higher caloric intakes (150–160 kcal/kg/day) due to tachypnea/increased work of breathing. Increase caloric density via fortification, provide frequent small feedings. Use nasogastric (NG) feeding if needed.



Older Children:

Strict bed rest, salt restriction, avoidance of salty snacks.

VI. PHARMACOLOGICAL THERAPY



A. Diuretics (Preload Reduction)

Principal agents for congestion, do not improve CO directly.

Loop Diuretics (Furosemide): Absolute drug of choice.



Side Effects: Hypokalemia (premeditates digitalis toxicity), hyponatremia, hypomagnesemia, hypovolemia, hypotension, ototoxicity.

Aldosterone Antagonists (Spironolactone): Added to loop diuretics to prevent hypokalemia & overcome resistance.

Discontinue if ACE inhibitors are used to avoid fatal hyperkalemia.

B. Afterload-Reducing Agents (Vasodilators)



Vital for infants with large Left-to-Right shunts (VSD, PDA) and DCM.

Mixed Vasodilators (ACE Inhibitors - Captopril, Enalapril): Act on both arteriolar and venous beds. Highly popular for chronic severe HF.

C. Inotropic Agents (Myocardial Performance)



Rapid-Acting Inotropes (Dopamine, Dobutamine, Epinephrine): Used strictly in critically/ acutely ill.

Phosphodiesterase Inhibitors (Milrinone): Augment contractility and decrease afterload with less chronotropy.

D. Digitalis Protocol (Digoxin)



Indications: HF with decreased systolic function (e.g., DCM) & atrial arrhythmias. Added later in large shunts.

Absolute Contraindications: Digitalis toxicity, paroxysmal ventricular tachycardia, Obstructive lesions (TOF, Hypertrophic Cardiomyopathy). Avoid co-administration with calcium, adrenaline, or vagolytic atropine.



Age-Based Oral Digitalizing Dosages:

Premature: \$20 mcg/kg **Newborn:** \$30 mcg/kg
< 2 years: \$40–50 mcg/kg **> 2 years:** 30–40 mcg/kg

Administration: details in fnovbn on: 30–40 mcg/kg.

Digitalis Toxicity Symptoms: Anorexia, nausea, vomiting, bradycardia, visual disturbances, severe arrhythmias.

Toxicity Management: Stop digoxin immediately, stop potassium-losing diuretics, give potassium supplements, treat arrhythmias, administer Digitalis specific antibodies.

E. Beta-Adrenergic Blockers (e.g., Carvedilol)



Carvedilol is drug of choice.

Used strictly in stable chronic stages of DCM (Class II–III).

Never use in acute phase; always combine with diuretics and ACE inhibitors.

CHAPTER 08

VIII. CNS



CEREBRAL PALSY • FLOPPY INFANT • DUCHENNE MUSCLE DYSTROPHY • INFECTION

CEREBRAL PALSY (CP): COMPREHENSIVE GUIDE

I. DEFINITION & CORE CHARACTERISTICS

- **Concept:** It is an encephalopathy resulting from malfunction of the motor unit of the developing brain due to a brain insult during prenatal, natal, and postnatal periods.
- **Core Deficit:** Central motor deficit that is strictly **non-progressive**, **non-familial**, and **non-hereditary**.
- **Associated Deficits:** May be associated with mental retardation, epilepsy, visual defect, deafness, and speech defect.



II. ETIOLOGY & TIMING OF INSULT

Prenatal Factors (80%): TORCH infection, teratogens, and intrauterine hypoxia.



Perinatal Factors (10%): TORCH infection, teratogens, intrauterine hypoxia.



Perinatal Factors (10%): Birth asphyxia, birth trauma, and IC hemorrhage.



Postnatal Factors (10%): Kernicterus, meningitis, encephalitis, and IC hemorrhage.

III. CLINICAL CLASSIFICATIONS

A. Topographic (Anatomic) Classification



Hemiplegia: Arm & Leg on same side of body (R or L).



Diplegia: Arms and Legs but legs are more involved.



Quadriplegia or Tetraplegia: Arms and Legs equally involved OR Arms more involved than legs.



Monoplegia: One limb.



Triplegia: Three limbs.



Paraplegia: Only legs involved.

B. Physiological Classification (Major Motor Abnormalities)

Spastic CP (70 to 80%): Hypertonia with hyperreflexia, positive Babinski sign (extensor plantar response), and persistent primitive reflexes.



Dyskinetic CP (10 to 20%): Abnormal movements and/or change in tone (increased or decreased).



- **Rigidity:** Hypertonia with hyporeflexia or normal reflexes.
- **Chorea:** Quick irregular dysrhythmic movements involve proximal skeletal muscles.
- **Athetosis:** Slow twisting movements, usually along long axis of a limb, more prominent distally.
- **Choreoathetosis:** Chorea and Athetosis are commonly seen.



Mixed Forms: Spasticity and ataxia may coexist.

- **Dystonia:** Painful sustained contraction of a muscle group's → prolonged, twisting, posturing of the face, limb or trunk, which involved.



Ataxic CP (5 to 10%): Disturbances in balance and equilibrium: walk unsteadily, poor coordination and balance often with hypotonia, hyporeflexia, nystagmus.



Atonic (Hypotonic) CP: Floppy infant with hyperreflexia, usually with marked mental retardation.



C. Functional Classification



Class 1: No limitation of activity.



Class 2: Slight to moderate limitation.



Class 3: Moderate to great limitation.



Class 4: No useful physical activity.

IV. MULTIDISCIPLINARY MANAGEMENT

The Team: A team of physicians: pediatricians, occupational and physical therapists, social workers, speech pathologists.



The Role of Pediatrician:



Treatment of comorbidities (seizures, malnutrition, spasticity).



Supervising nutritional status.



To prescribe muscle relaxants (for muscle spasticity), sedatives (to reduce sleep and decrease irritability).



To explain with the help of the social workers and psychiatrist the nature of the disease to the family to ensure better cooperation.



Refer to orthopedic surgery to correct deformities, provide stability and rehabilitation.



THE FLOPPY INFANT SYNDROME

I. CORE DEFINITION

Concept: Severe persistent hypotonia present at birth or early infancy.



II. ANATOMICAL & ETIOLOGICAL CAUSES



A. Cerebral Causes



Perinatal Asphyxia: Brain insult leading to central hypotonia.



Kernicterus: Bilirubin-induced encephalopathy.



Cerebral Palsy: Specifically the atonic (hypotonic) type.



Chromosomal Anomalies: Including Down syndrome.



B. Spinal Cord Lesions



Anterior Horn Cell Disease: Werdnig-Hoffman spinal muscular atrophy.



C. Peripheral Nerves



Acute Polyneuropathy: Generalized dysfunction of peripheral nerves.



Guillain-Barré Syndrome (GBS): Post-infectious demyelinating polyneuropathy.



Congenital Sensory Neuropathy: Inherited structural or functional nerve abnormalities.



B. Spinal Cord Lesions



Anterior Horn Cell Disease: Werdnig-Hoffman spinal muscular atrophy.



Early Extensive Poliomyelitis: Viral destruction of lower motor neurons.



D. Neuromuscular Junction



Neonatal Myasthenia Gravis: Transient autoimmune neuromuscular blockade.



Infantile Botulism: Neurotoxin-induced inhibition of acetylcholine release.



E. Myopathies



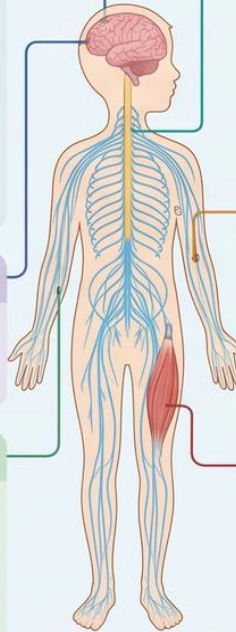
Muscular Dystrophies: Progressive genetic muscle wasting disorders.



Congenital Myopathies: Structural embryonic muscle abnormalities.



Polymyositis: Inflammatory muscle disease.



III. CLINICAL SIGNS & EVALUATION OF HYPOTONIA



Frog Leg Position: Denotes marked hypotonia of the lower limbs.



Head Lag: When the baby is pulled up from his hands while in a supine position, the head lags backwards, denoting hypotonia of the neck muscles.



Curved Trunk (Ventral Suspension): When the baby is suspended in a prone position over the examiner's hand, they drop around it in a "U-shape", denoting hypotonia of the trunk muscles.



Slippage (Vertical Suspension): The infant slips through the examiner's hands when held under the arms, denoting hypotonia of the shoulder girdle muscles.

IV. DIAGNOSTIC DIFFERENTIATION (CENTRAL VS. PERIPHERAL)



A. Cerebral Hypotonia Criteria



Abnormal Brain Functions: Presence of seizures or a disturbed conscious level.



Cognitive Deficit: Association with mental retardation.



Dysmorphic Features: Phenotypic signs of chromosomal disorders (e.g., Down syndrome).



Deep Tendon Reflexes: Characterized by hyperreflexia.



B. Lower Motor Unit Disorders Criteria



Deep Tendon Reflexes: Characterized by areflexia or hyporeflexia.



Muscle Bulk: Prominent muscle atrophy and fasciculations.



Neuropathy Pattern: Total loss of reflexes with residual movements.



Myopathy Pattern: Diminished reflexes that are directly consistent with the degree of muscle weakness.

Diagnostic Differentiation Test



These lower motor unit conditions are definitively differentiated by performing Nerve Conduction Velocity (NCV) and Electromyography (EMG).



[Image Missing: Duchenne muscle dystrophy.jpeg]

[Image Missing: Infections.jpeg]